



CONTACT

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TECHNICAL SKILLS

- Python, SQL, pandas, NumPy
- scikit-learn, Machine Learning
- Power BI, DAX, Excel
- NLP, TF-IDF, Recommendations
- Streamlit, FastAPI, Docker
- Git, GitHub, REST APIs
- C#, JavaScript, PHP, MySQL

BUSINESS ANALYSIS

- Requirements elicitation
- Stakeholder analysis
- Process and workflow modelling
- Use cases, user stories
- Documentation and reporting

SOFT SKILLS

- Communication
- Leadership
- Teamwork
- Critical Thinking
- Problem Solving
- Analytical Thinking
- Adaptability

LANGUAGES

- Sinhala - Fluent
- English - Fluent (C1)
- Japanese - Intermediate

RUVIN PALIHAWADANA

DATA SCIENCE | BUSINESS ANALYSIS | MACHINE LEARNING

PROFILE

Business Information Systems and Applied Data Analytics undergraduate with hands on experience in machine learning, NLP, recommendation systems, data analysis, requirements engineering, process modelling, and full-stack application deployment. Built practical solutions using Python, SQL, scikit-learn, Power BI, Streamlit, FastAPI, Docker, and web technologies. Strong business and accounting foundation with proven analytical, communication, and project-coordination skills.

EDUCATION

BSc (Hons) in Business Information Systems	2024 - Present
• University of Sri Jayewardenepura • Department of Information Technology • Faculty of Management Studies and Commerce. • Relevant study: Business Analytics, Systems Analysis, Python/C# Programming, Database Management, Research Methodology.	
BSc (External) in Applied Data Analytics	In Progress
• University of Colombo • focus on statistics, Python, R, SQL, machine learning, data visualization, text mining, and artificial intelligence.	
Higher National Diploma in Information Technology (HND in IT)	In Progress
• University of Moratuwa.	
GCE Advanced Level - Commerce Stream	2022/2023
• Taxila Central College, Horana - District Rank 26.	

PROFESSIONAL CERTIFICATIONS

IBM Data Science Professional Certificate - IBM, Coursera | 2026
Supervised Machine Learning: Regression and Classification - DeepLearning.AI & Stanford Online | 2026
Google AI Essentials Specialization - Google, Coursera | 2026
Data Science with Python - Udemy | August 2025
10 Python Data Science Projects - Udemy
Certificate in Advanced English (C1) - Cambridge English | 2023-2024

DATA SCIENCE & AI PROJECTS

SOLVEMATE - Polymer Solvent Compatibility Prediction System

Python, Random Forest, Fast API, Docker, Railway

- Developed a machine learning classifier using approximately 27,050 polymer-solvent records, achieving 84.94% accuracy and 0.9036 ROC-AUC.
- Deployed the model as a FastAPI service integrated with a Spring Boot backend and Vite frontend.

Multilingual Scam Detection System

Python, NLP, text and URL-risk features

- Developed an NLP-based system to classify Sinhala, Tamil, English, and Singlish messages as Scam, Suspicious, or Ham.
- Combined text-based indicators with URL-risk features to provide risk-level guidance.

CinePulse - Emotion-Aware Movie Recommendation

TF-IDF, cosine similarity, NRCLex, Streamlit, TMDB

- Built an emotion- and content-based movie recommendation application using TF-IDF, cosine similarity, and emotion analysis.
- Deployed the application through Streamlit with TMDB movie-poster integration.

Dengue & Socioeconomic Factor Analysis

Python, feature importance, LOOCV

- Integrated district-level dengue, population density, poverty, income, and Samurdhi data for Galle, Matara, and Hambantota.
- Applied feature-importance analysis and Leave-One-Out Cross-Validation to examine socioeconomic factors associated with dengue cases.

SLM-Based RAG Security Framework

Research in progress

- Designing lightweight prompt-injection and data-leakage detection across prompts, retrieved context, and generated responses.

Support Bot – Transformer-Based Desktop Assistant

Python, Transformers, PyQt6, gTTS, Text-to-Speech

- Developed a desktop-based support assistant with a PyQt6 graphical interface and Transformer-based language processing.
- Integrated gTTS and text-to-speech functionality to provide spoken responses and improve user accessibility.

Customer Churn Prediction System

Python, Machine Learning, Feature Engineering, Classification

- Developed a classification-based system to identify customers who may be at risk of leaving a service.
- Performed data preprocessing, feature engineering, model training, and evaluation, and created a simple interface for generating predictions

BUSINESS ANALYSIS & SOFTWARE PROJECTS

Debit Card Renewal Management System and Business Analysis Report

PHP, JavaScript, MySQL, requirements engineering, Process Modelling

- Designed a digital debit-card renewal workflow covering card-expiry alerts, renewal requests, KYC coordination, application-status tracking, and delivery follow-up.
- Prepared comprehensive business-analysis documentation, including project scope, stakeholder analysis, functional and non-functional requirements, use cases, user stories, acceptance criteria, process workflows, database design, testing, and proposed improvements.

BookInn – Hotel Management System and Business Analysis Report

C#, WPF, SQL, process analysis

- Analysed manual hotel operations and designed an integrated system for room availability, guest registration, reservations, check-in and checkout, billing, payments, and management reporting.
- Produced requirements documentation, process flowcharts, pseudocode, database designs, testing considerations, and workflow-improvement recommendations..

Craftpolio - AI Portfolio Generator

AI-Assisted Document Processing, Resume Extraction, Web Development

- Developed an AI-assisted application that extracts information from resumes and generates personalized portfolio website content.
- Automated the portfolio-creation workflow and enabled users to preview and download the generated website output.

GoAnime – Anime Discovery Website

HTML, CSS, JavaScript, AniList GraphQL API

- Developed a responsive anime-discovery website that retrieves and presents anime information through the AniList GraphQL API.
- Implemented search and browsing functionality to help users explore anime titles, descriptions, ratings, genres, and related details.

Python and PyGame Game Development

Python, PyGame, Game Logic, Collision Detection, Audio Integration

- Developed multiple arcade-style games, including **SpaceWars**, **Brainiac Attack**, and **a Robot Wars**
- Implemented player movement, enemy spawning, shooting mechanics, collision detection, scoring, sound effects, background audio, and game-state management.

PROFESSIONAL AND VOLUNTEER EXPERIENCE

IGV B2B Member – AIESEC in University of Sri Jayewardenepura

Volunteer | 2025–2026

Supports stakeholder communication, outreach, partnerships, and project execution

WORKING EXPERIENCE

Assistant Paper Marker - SARVA

2023-2024

Maintained accuracy, consistency, confidentiality, and examination quality control.

EXTRA-CURRICULAR ACTIVITIES

OCVP - Logistics, TRIBUTE, AIESEC | 2025/26: Led logistics planning and coordinated venues, equipment, resources, schedules, vendors, and on-ground event operations to support successful event execution

Logistics Operations Manager - ISACA Student Group | 2025/26: Coordinates resources, schedules, and on-ground event operations.

Finance Committee Member - Department of IT | 2025/26: Supports budgeting, expense tracking, sponsorships, vendor payments, and financial records.

AWARDS & RECOGNITION

Prize Winner - Management Accounting (BL6), CA Sri Lanka | 2023.

Second Runner-Up - CRYPTX 2.0 Designathon, University Category | 2026.

AIESEC Recognition. Top-performing member and Best Content Writer nominee.

REFERENCES

U. S. Samarasinghe

Lecturer (Probationary)

Department of Information Technology, Faculty of Management Studies and Commerce

University of Sri Jayewardenepura

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